TEXORA · LECTURE SHEET

Pile Fabrics

A pile fabric grows a third dimension out of the ground cloth: loops or cut fibre ends stand up from a plain or twill ground, put there by an extra warp (velvet, terry) or an extra weft (velveteen, corduroy). This sheet compares the two families construction by construction, down to the design grid of a weft-pile ground.

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1. The idea: ground + pile

Every pile fabric is a **ground weave** (plain or twill, which gives the cloth its strength) plus a **pile system** — additional yarn that is bound into the ground at intervals and elsewhere stands free of it as a **loop** (uncut pile) or as a **cut end** (cut pile). The pile system is classified by which yarn supplies it:

FAMILY	PILE YARN	MAIN MEMBERS	PILE FORMED BY
Warp pile	Extra warp ends, from a second warp beam	Velvet, plush, terry towelling	Pile warp floats over wires or rods, or is beaten up slack; loops cut (velvet) or left (terry)
Weft pile	Extra picks, in a high pick density	Velveteen, corduroy	Pile picks float over ground ends; the floats are cut afterwards along their length

Two quantities run through everything: **pile ratio / density** (how much pile yarn there is relative to ground — for warp pile, the arrangement ground : pile, e.g. 2 ground ends : 1 pile end, and the length ratio of pile-warp feed to ground-warp feed) and **pile height** (wire diameter for velvet, float length and cutting for weft pile, loop feed length for terry).

2. Warp pile I — velvet and plush (over-wire method)

The pile warp is interlaced into the ground in a fast (firm) binding — typically the pile end binds in a **W or V pattern** over ground picks — and between bindings it is lifted over a **wire** inserted like a pick across the warp. When the wire is withdrawn it leaves a loop; wires carrying a knife edge cut the loop as they are drawn out, giving **cut pile (velvet)**; plain wires give loop pile (plush when the loops are long). Two cloths can also be woven **face to face** with the pile warp travelling between them and cut apart as they leave the loom — the double-velvet method, which needs no wires and produces two velvets at once. Pile height = wire height (or half the layer gap, face-to-face); the ground:pile end arrangement is commonly 2:1 or 3:1, finer ratios giving a denser, more expensive pile surface.

3. Warp pile II — terry (loop pile by slack tension)

Terry towelling forms its loops without wires: the ground warp is held taut, the **pile warp is fed slack**, and a special **loose beat-up (terry motion)** works in a 3-pick cycle — two picks are beaten up only part way (short beat-up, left standing off the cloth fell), then the third pick is driven fully home; the group of three picks slides along the taut ground warp and pushes the slack pile yarn up into **loops on one or both faces**. The **pile ratio** (pile-warp length consumed ÷ ground-warp length) is typically about **3:1 to 6:1** and directly sets loop height and towel weight/absorbency: at 4:1, every metre of ground consumes four metres of pile warp. Loops may be left on both faces (double-sided terry) or one face only. The examination trap: terry loops are formed by *beat-up mechanics and differential warp tension*, not by the weave design alone — the design merely binds the pile end in plain order.

4. Weft pile — velveteen and corduroy

Here the ground is a normal warp with ground picks, and the **pile picks** — inserted in a high ratio, commonly 2–4 pile picks to 1 ground pick — float over several ground ends and bind at single points. After weaving, the floats are **cut along their centres** (by rotating disc knives guided along the float lines), and the cut ends brush up into pile. The arrangement of the binding points decides the fabric:

- **Velveteen:** binding points distributed all over (plain or twill ground, short floats in a staggered order) → an even, all-over short pile with no direction lines.
- **Corduroy:** binding points aligned in columns, so the floats form continuous lengthwise channels; cutting opens ribs (wales) of pile separated by ground lines. Corduroy is graded by **wales per inch**: fine **needlecord/pincord ≈ 14–16**, standard ≈ 11, wide/jumbocord ≈ 3–8.

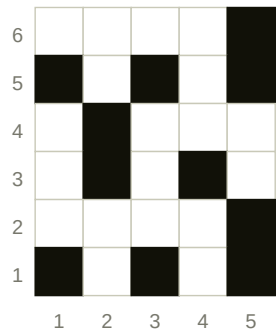


Figure 1 — Weft-pile design (one repeat: 5 ends × 6 picks). Picks alternate ground (G, plain weave) and pile (P): on each pile pick the warp rises at only ONE binding end (staggered: end 5, then end 2, then end 5), so the pile pick floats across the other **4 ends** — those floats, cut along their centres after weaving, become the pile tufts. Verify: every pile-pick row has exactly 1 mark; the ground rows keep perfect plain order.

Why the float is short (4–5 ends) in velveteen-type designs: the cut pile tuft is held only by its single binding interlacing; longer floats give longer pile but pull out in wear. Corduroy's floats run along the rib for the same reason wale by wale. After cutting, the cloth is singed, brushed and sheared to even the

pile — the finishing sequence that makes the difference between a grey-state curiosity and velvet's level surface.

5. The two families compared

FEATURE	WARP PILE (VELVET, TERRY)	WEFT PILE (VELVETEEN, CORDUROY)
Pile yarn supply	Second warp beam; pile ratio set by feed	Extra picks; pile ratio set by pick plan
Loop/cut mechanism	Wires, face-to-face cutting, or slack beat-up (terry)	Floats cut after weaving along their length
Pile direction	Loops/ends stand along the warp	Tufts from weft floats; ribs run warp-way
Typical cost driver	Pile-warp consumption (ratio 3–6 in terry)	Very high pick densities; cutting and finishing
Failure mode	Pile shedding at cut edges; loop pulling (terry)	Pile pull-out from single bindings; rib crushing

Glossary

Pile — loops or cut ends standing from a ground weave. **Ground weave** — the base interlacing giving strength. **Pile ratio** — feed length (or thread proportion) of pile system relative to ground. **Wale (corduroy)** — one lengthwise rib of cut weft pile; graded per inch. **Terry motion** — loose beat-up cycle that forms terry loops. **Fast pile** — pile binding woven firmly into the ground (W/V binding).

Key points

• Pile = ground + an extra system; classify by whether the extra system is warp or weft.
• Velvet: pile warp over wires or face-to-face, then cut. Terry: slack pile warp + 3-pick loose beat-up; ratio ~3:1–6:1 sets loop height.
• Weft pile: pile picks float (4 ends in Figure 1) and bind once; staggering the bindings makes velveteen, aligning them makes corduroy.

Practice

1. A terry warp is arranged 2 ground : 1 pile and the pile ratio is 4:1. For 1 m of ground warp consumed, how much pile warp is consumed, and what does the ratio control?

2. In Figure 1, the pile pick of pick 4 binds at which end, and what is its float length?
3. A buyer wants an all-over even pile with no ribs, in cotton, woven as weft pile. Name the fabric and the binding-point arrangement it needs.
4. Why is terry never described as "a design with loops in it"? State the two loom conditions that actually form the loop.

Answer key

1. 4 m of pile warp per 1 m of ground warp. The ratio controls loop height (and with pile density, the towel's weight and absorbency): higher ratio, longer loops fed per repeat.
2. Pick 4 is the second pile pick; it binds at **end 2** (the stagger: ends 5, 2, 5), floating over the other **4 ends**.
3. **Velveteen**: pile-pick binding points distributed evenly all over the repeat in a staggered order (never aligned in columns), so the cut pile covers the surface uniformly with no channels.
4. Because the loop is formed mechanically, not by interlacing geometry: (i) the pile warp is delivered slack (low tension, high feed ratio) against a taut ground warp, and (ii) the terry (loose beat-up) motion drives a group of picks home together so they slide on the ground warp and throw the slack pile yarn up into loops. The design only binds the pile end in plain order.

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